THE SPECTRAL SEQUENCE OF AN AUGMENTED SIMPLICIAL SITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: in-source stubs lemma-augmentation-cech-complex, lemma-restriction-to-components-site, lemma-restriction-injective-to-component-site; external tools from Homology.

Editorial note. The label lemma-augmentation-spectral-sequence names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Augmentation Cech complex. Same unit, Lemma lemma-augmentation-cech-complex. Provides the horizontal arrows of the double complex (rows exact, evaluating to a_*I^q in degree 0).
- Exactness of restriction. Same unit, Lemma lemma-restriction-to-components-site. Restriction to C_p exact, so I_p^bullet represents K_p.
- Injectivity of restrictions. Same unit, Lemma lemma-restriction-injective-to-component-site. I_p^q injective on C_p, so column cohomology computes R^q a_{p,*} K_p.

Source reference. Original lemma 0133: sources/stacks-chow/tags.txt, source lines 1421–1421. Prerequisite pointer invoked at proof line(s) 2054; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0132: sources/stacks-chow/tags.txt, source lines 1420–1420. Prerequisite pointer invoked at proof line(s) 2053; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D79: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1988–1988. Prerequisite pointer invoked at proof line(s) 2041; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 09WH: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1086–1086. Prerequisite pointer invoked at proof line(s) 2050; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 2046; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6Z: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 1125–1125. Prerequisite pointer invoked at proof line(s) 2023; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at proof line(s) 2021; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let a_0 be an augmentation towards a site $\mathcal{D}$ as in Remark 0D6Z. For any K in $D^+(\mathcal{C}_{total})$ there is a spectral sequence $(E_r, d_r)_{r\geq 0}$ with*

$$E_1^{p,q} = R^q a_{p,*} K_p, \quad d_1^{p,q} : E_1^{p,q} \to E_1^{p+1,q}$$

converging to $R^{p+q}a_*K$. This spectral sequence is functorial in K.

Proof. Let $\mathcal{I}^\bullet$ be a bounded below complex of injectives representing K. Consider the double complex with terms

$$A^{p,q} = a_{p,*}\mathcal{I}_p^q$$

where the horizontal arrows come from Lemma 0D79 and the vertical arrows from the differentials of the complex $\mathcal{I}^\bullet$. The rows of the double complex are exact in positive degrees and evaluate to $a_*\mathcal{I}^q$ in degree 0. On the other hand, since restriction to $\mathcal{C}_p$ is exact (Lemma 09WG) the complex $\mathcal{I}_p^\bullet$ represents K_p in $D(\mathcal{C}_p)$. The sheaves $\mathcal{I}_p^q$ are injective abelian sheaves on $\mathcal{C}_p$ (Lemma 09WH). Hence the cohomology of the columns computes $R^q a_{p,*} K_p$. We conclude by applying Homology, Lemmas 0132 and 0133. □